

# Shantinath Patil

## AI/ML Engineer

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### Summary

AI/ML Engineer with a peer-reviewed RAG publication and three deployed end-to-end projects spanning RAG pipelines, supervised learning, and clustering. Proficient in Python, LangChain, Hugging Face, and vector databases, with hands-on experience shipping ML applications from data prep through deployment.

### Skills

|                      |                                                                                                           |
|----------------------|-----------------------------------------------------------------------------------------------------------|
| Languages            | Python, C                                                                                                 |
| Libraries/Frameworks | NumPy, Pandas, Scikit-learn, TensorFlow, LangChain, Hugging Face, FastAPI, Streamlit, Matplotlib, Seaborn |
| Core Areas           | Machine Learning, Deep Learning, NLP, Generative AI, RAG, EDA, Feature Engineering, Model Evaluation      |
| Databases            | MySQL, MongoDB, ChromaDB                                                                                  |
| Tools                | Git, GitHub, Jupyter, Google Colab, VS Code, Postman, MySQL Workbench, Render, Vercel                     |

### Experience

**Machine Learning Intern at FlyRank AI** Sep 2026 – Present

- Onboarded onto the team's ML workflows and tooling; currently supporting ongoing model development and data pipeline tasks. *(Update with a specific task/tool once assigned.)*

**NLP Intern at CCC Innovation and Incubation Center** Jan 2024 – Jun 2024

- Built an NLP-based chatbot using intent classification and dialogue-flow design to enable automated, context-aware conversations.
- Improved response relevance by refining natural language understanding logic and testing against real conversation scenarios.
- Collaborated with mentors to scope the chatbot's use case and validate its flows against real user queries.

**Student Ambassador at Google Gemini** May 2026 – Present

- Lead hands-on workshops and awareness sessions on Generative AI tools for the student community, driving practical adoption of Google Gemini.
- Serve as a peer point of contact for Gemini-related queries within the student community.

### Projects

**RAG System – Local Retrieval-Augmented Generation** GitHub

- Built a fully local RAG pipeline (LangChain + ChromaDB) for document-grounded Q&A with zero external API dependency, cutting inference cost and latency vs. API-based alternatives.
- Designed the chunking, embedding, and retrieval flow to ground LLM responses in source documents and reduce hallucinations. *Architecture:* Chunking → Embeddings → ChromaDB → Retrieval → LLM Response.

**Mental Health Signal – Student Wellness Analytics** Live Demo · GitHub

- Trained a supervised ML model on a dataset of 50,000 student records to predict wellness scores from lifestyle and academic indicators, achieving 87% accuracy; deployed as a live web app on Render.
- Ran EDA and feature engineering to identify the strongest predictors of student wellbeing, informing feature selection for the final model.

**Smart Clustering System** Live Demo · GitHub

- Built an interactive Streamlit app supporting multiple clustering algorithms with user-adjustable parameters and real-time visualization of cluster formation.

### Publications

**Large Language Models Hallucinate and How Retrieval-Augmented Generation Mitigates It** – IJRASET, Vol. 14, Issue VIII, Aug 2026 (Paper ID: IJRASET84752). [Link](#)

### Education

**Bahubali College of Engineering, Shravanabelagola** Jan 2023 – Dec 2027  
*B.E., Artificial Intelligence and Machine Learning* CGPA: 8.80

### Certifications

AI/ML – Apna College (2026) · ISRO – Geo Data Sharing and Cybersecurity · Certified LLM Security Professional · 5-Day AI Agents Intensive Course – Google